From Algorithms to Agents: A Survey of LLM-Based Trading Systems

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Large language models (LLMs) now drive trading systems, from signal extractors to autonomous agents, but the literature is fragmented across markets, architectures, and evaluation practices. This survey traces the path from quantitative trading to LLM agents, organizes the literature by capability (perception, reasoning, action, memory, collaboration) and by six markets from equities to prediction markets, and maps 47 systems across both to show where evidence is thin. We audit evaluation protocols and consolidate head-to-head evidence against traditional quantitative baselines. Four findings emerge. First, the margins of independently re-run agents shrink or vanish, though the re-runs cover few systems and many reported margins rest on windows of eight months or less, where an annualized Sharpe ratio has a standard error above one. Second, the best-supported hypothesis places the LLM upstream of the trade (text signals, factor and model code, and bounded execution tilts), although most upstream evidence comes from single author-run studies. Third, controlled comparisons do not establish a multi-agent advantage, though risk controls help where ablated. Fourth, structure and input design matter alongside the backbone, and the best structure depends on the backbone. Open challenges span reliability, security, deployment, regulation, market stability, reproducibility, and human-AI collaboration.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Multi-agent systems**; • **Information systems** → *Data mining*.

Additional Key Words and Phrases: Large Language Models, Autonomous Agents, Financial Trading, Quantitative Trading, Multi-Agent Systems, Cryptocurrency, Survey

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ACM 1557-7341/2026/10-ART0
https://doi.org/10.1145/nnnnnnn.nnnnnnn

ACM Reference Format:
Yebo Feng, Yichen Luo, Jingyang Dai, Chien-Chih Chen, Guangyan Gan, Jiahua Xu, Lin William Cong, and Yang Liu. 2026. From Algorithms to Agents: A Survey of LLM-Based Trading Systems. *ACM Comput. Surv.* 0, 0, Article 0 (October 2026), 35 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Trading systems have moved from rule-based strategies and statistical arbitrage through deep learning and reinforcement learning (RL) [61, 121, 137] to large language models (LLMs), which now power signal extractors and agents that trade with diminishing human oversight [169, 183, 188]. LLM agents read text [106] and chart images [112], reason into natural-language theses [159, 168], and organize as multi-agent firms that reportedly surpass single-agent [183] and rule-based [169] baselines.

The topic matters now because deployment is outpacing evidence. In a survey by the International Organization of Securities Commissions (IOSCO), 63.3% of industry respondents report using AI for algorithmic trading [66]. Supervisors expect autonomous AI to move quickly into trading and warn that agents responding alike could amplify volatility in stress [15, 42]. The research that should inform these decisions is hard to trust: in 164 main-conference papers, no finance-specific evaluation bias is discussed by more than 28% [78], and an audit of 185 studies finds protocol reporting improving but still incomplete [167]. Practitioners and regulators need a synthesis that separates reported from reproducible performance.

Existing surveys do not provide it. Since late 2022, LLM trading papers have proliferated [32, 94, 134, 170, 183, 188], but surveys cover LLMs in finance [82, 95, 119], LLM agents generally [158, 166], or, concurrently, LLM trading agents [19, 29, 32]; none combines broad market coverage, a capability-centric agent taxonomy, and systematic cross-paradigm comparison. Drawing on 246 works, 45 of them primary studies in our evidence tables, we make three contributions:

- **Paradigm-transition analysis.** We trace the move from quantitative trading to LLM agents, showing what is new and where traditional methods remain competitive.
- **Capability × market map.** We cross a capability-centric taxonomy, adapted from general agent frameworks [144, 158], with six markets from equities to cryptocurrency and decentralized finance (DeFi), forex, derivatives, and prediction markets, and code 47 systems into the resulting map (Figure 3). Its empty cells locate the gaps: every memory design comes from an equity trader, and the other markets rest mostly on few systems or on bounded LLM components.
- **Evidence audit and cross-paradigm comparison.** We set reported results against same-window passive baselines, audit their protocols, grade multi-agent claims by how well they control confounds, examine why author-run (self-evaluated) and independent results diverge, and consolidate 25 head-to-head studies against quantitative methods.

The audit supports four findings: independently re-run margins shrink or vanish (§7.4); the best-supported current hypothesis places the LLM upstream of the trade, in text signals, factor and model code, and bounded execution tilts (§8.6); a multi-agent advantage is not established (§4.3); and structure and input design matter alongside the backbone, with the best structure depending on the backbone (§4.3, §5.2). Our evidence tables are purposive and over-represent highly cited 2023–2024 systems, and 20 of the 45 primary studies are preprints (Supplement §S11). An online supplement gives a glossary, the methodology, and the full evidence tables.

2 Related Surveys

Table 1 scores coverage overlap; the surveys fall into three lines, each missing a different part of the problem. Pre-LLM surveys of machine, deep, and reinforcement learning for trading [56, 61,

Table 1. Comparison with related surveys (○: not included; ◐: partially included; ●: included). #Refs: number of references; †number of included studies instead, as the reference count was not determined. The six subject dimensions map to core sections of this survey: Paradigm Transition (§3), Agent Architecture (§4), Multi-Agent (§4.3), Multi-Market Applications (§6), Evaluation & Benchmarks (§7), and Cross-Paradigm Comparison (§8). Because the dimensions are this survey's organizing axes, the table shows overlap of coverage, not quality; §2.1 states the substantive differences.

Survey	Year	#Refs	Description	Subjects Covered					
				Paradigm Transition	Agent Architecture	Multi-Agent	Multi-Market Applications	Evaluation & Benchmarks	Cross-Paradigm Comparison
Traditional ML/DL Trading Surveys									
Ozbayoglu et al. [121]	2020	202	Deep learning (CNN, RNN, LSTM) for stock trading, portfolio management, and risk.	◐	○	○	◐	◐	○
Sezer et al. [137]	2020	220	Systematic review of deep learning for financial time series forecasting, 2005–2019.	◐	○	○	◐	●	○
Cao [20]	2023	221	AI in finance: classic statistical methods versus modern deep learning across tasks.	●	○	○	◐	◐	◐
Hambly et al. [56]	2023	246	RL in mathematical finance: execution, portfolio optimization, option pricing.	◐	○	○	◐	◐	○
Sun et al. [145]	2023	165	RL for quantitative trading: algorithmic trading, portfolio management, execution, market making.	◐	○	○	◐	●	○
Pippas et al. [127]	2025	356	Evolution of RL in quantitative finance, incl. transfer, meta-, and multi-agent RL.	●	○	◐	◐	●	◐
LLM-Based Agent Surveys									
Xi et al. [166]	2025	675	LLM agent framework (brain, perception, action) with multi-agent interaction across domains.	○	●	●	○	◐	○
Guo et al. [53]	2024	88	Multi-agent LLM systems: communication protocols, role specialization, coordination.	○	●	●	○	◐	○
Sumers et al. [144]	2024	235	CoALA: agent design grounded in cognitive science (memory, action, decision).	○	●	○	○	○	○
Wang et al. [158]	2024	185	Unified four-module agent framework (profiling, memory, planning, action).	○	●	◐	○	◐	○
LLM Agents for Finance Surveys									
Li et al. [95]	2023	68	LLM adoption guide for finance with a four-level framework (zero-shot to pre-training) and costs.	◐	○	○	○	◐	○
Lee et al. [82]	2025	54	Evolution from pre-trained language models to financial LLMs; five training techniques compared.	◐	○	○	○	●	○
Nie et al. [119]	2024	318	LLM application areas in finance: linguistic tasks, sentiment, time series, reasoning, agent-based modeling.	◐	◐	◐	◐	●	○
Ding et al. [29]	2026	60	First LLM trading-agent survey; "Trader vs. Alpha Miner" taxonomy; equities only.	○	●	◐	○	◐	○
Dong et al. [32]	2025	68	Maps LLM agents to five financial-institution divisions with a benchmark catalog.	○	●	●	◐	●	○
Cao et al. [19]	2025	186	DL-to-LLM evolution along the alpha pipeline (data, prediction, optimization, execution).	●	◐	◐	○	◐	●
Mahdavi et al. [113]	2025	—	LLM-for-investment work in four technique buckets; descriptive, equity-centric.	○	◐	◐	○	○	○
Fu [45]	2025	—	Task-centered survey of LLMs for equity return prediction and trading; time-safe, economically grounded evaluation.	◐	◐	○	○	●	◐
Gong [48]	2026	—	Conceptual "agentic finance" framework and systemic-risk model of agent markets.	◐	◐	○	◐	○	○
Nguyen & Pham [118]	2026	—	Workshop survey of 12 multi-agent trading systems: design taxonomy, five evaluation failures, coordination-primacy hypothesis.	○	●	●	○	●	○
Xia et al. [167]	2026	185†	Audit of 185 studies (80 closed-loop) to Sep. 2026: protocol coverage, reproducibility tiers, reporting checklist.	◐	●	●	◐	●	◐
Ours	**2026**	**246**	**Capability + scenario dual-dimensional LLM trading agent survey with cross-paradigm comparison.**	●	●	●	●	●	●

121, 127] supply the baselines LLM agents must beat but no agent architecture. General agent surveys [144, 158, 166] contribute the module decompositions our taxonomy adapts, yet assume domain-agnostic environments without adversarial participants, regulation, or real-time execution. Finance surveys either address LLMs and financial NLP rather than trading systems [82, 95, 119], or review trading agents with limited market breadth: Ding et al.'s Trader/Alpha Miner taxonomy [29] and Cao et al.'s review [19] cover equities only, and Mahdavi et al. [113] restate self-reported returns without discussing contamination or comparability. The newest surveys audit rather than catalog: Xia et al. [167] assess protocol reporting across 185 studies and propose a reporting checklist but decline to compare performance, and Nguyen and Pham [118] hypothesize, without testing it, that coordination protocol matters more than backbone.

2.1 Positioning of This Survey

Our contributions (§1) differ from these surveys in four respects rather than in coverage of Table 1's dimensions: (i) market breadth beyond equities, with the capability × market map (§4.1) and each market's evidence gaps (§6); (ii) evidence grading of multi-agent claims, separating comparisons with published systems, within-system ablations, and matched single-agent controls (§4.3); (iii) a consolidation of independent re-evaluations that sets author-run margins against re-run ones (§7); and (iv) a consolidation of head-to-head evidence against traditional baselines that keeps signal-level and agent-level studies apart (§8).

3 Background: Paradigm Evolution in Quantitative Trading

Quantitative trading has passed through four paradigms (Figure 1), each relaxing one binding constraint. Supplement §S10 defines the finance and ML terms used throughout.

3.1 Traditional Quantitative Trading

Traditional quantitative trading lets humans specify signals and statistical models estimate them [20], through rule-based strategies [16], statistical arbitrage [46], and factor models from CAPM [139] to Fama-French [36]. Its binding constraint was human feature design, and even text entered this way, as dictionary sentiment scores built on a fixed vocabulary [108, 148].

3.2 Machine Learning and Deep Learning

Machine learning and deep learning moved feature construction to the model [51, 121, 137] but not past the low signal-to-noise ratio of financial data, and capacity did not guarantee gains: Feng et al.'s selected deep factor models reach out-of-sample Sharpe ratios of 2.49 and 3.00, below the 3.67 of a five-factor instrumented principal component analysis (IPCA) model [39]. Attention, the architecture behind LLMs, is likewise not automatically superior on numerical sequences; on limit-order-book prediction, LSTM variants beat Informer, Autoformer, and FEDformer on price-change and direction tasks [12]. Evaluation reported movement accuracy on disparate data sets rather than profitability under common protocols [194], a fragmentation that LLM agent studies inherited (§7.5). Reinforcement learning made the position the output [56, 145], and FinRL standardized environments [102], but sample efficiency and the sim-to-real gap remain open. Learned encoders such as FinBERT [5, 64] replaced text dictionaries but were each trained for one labeled target.

3.3 Large Language Models

LLMs made unstructured text a trading input through zero-shot reading of filings and news [165], chain-of-thought reasoning [163], in-context learning [17], and code generation. Yet the most cited early results cast the LLM as analyst or signal generator, from headline sentiment trading [106]

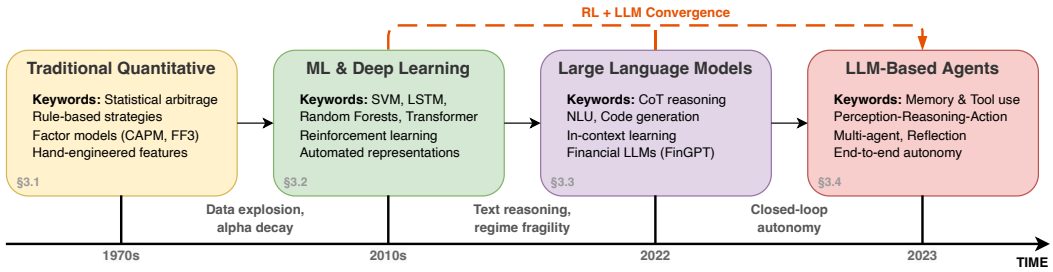


Fig. 1. Paradigm evolution in quantitative trading across four eras, with representative methods and the drivers of each transition.

to alpha mining [161], because a standalone LLM is stateless, passive, and disconnected from live data and execution.

3.4 From LLM to Agent

Agents close this loop by embedding the LLM in an architecture of perception, reasoning, action [179], memory [182], and multi-agent collaboration [169, 183], the five dimensions of our taxonomy (§4). Each transition changed one component: ML automated features and RL replaced prediction error with sequential reward, LLMs widened the information set to text, and agents widened the action space to tool calls, retrieval, and memory updates. What stayed fixed are the environment and its testing: noisy, non-stationary data, regimes that break fitted models, and evaluation by bespoke backtests. §7.5 and §8 test whether text, reasoning, and memory overcome these constants or merely add ways to overfit them.

4 Architecture of LLM Trading Agents

Five capability dimensions frame LLM trading agents: perception, reasoning, action, and memory within one agent (§4.1), and collaboration among several (§4.3). Three functional paradigms, Trader, Alpha Miner, and bounded component, place the LLM at different points of the pipeline (§4.2).

4.1 Overall Architecture and Information Flow

An LLM agent is a language model in a closed perceive–reason–act loop with memory and tools [141, 144, 179]. Trading keeps this loop but changes its constraints: perception fuses narratives with numerical series [106, 188]; actions are irreversible and subject to slippage, latency, and market impact [18, 123]; memory must survive regime shifts [182, 183]; and reasoning is judged on risk-adjusted performance, not task completion [10, 172]. The reasoning core is a prompted frontier LLM, a finance-specialized LLM [165, 170], or a smaller open-weight model trained end-to-end for trading [168, 193]. Four modules surround this core (Figure 2), each with an open challenge (techniques in §5).

- **Perception** treats text, images, and on-chain data as first-class signals, and retrieval decides what reaches the reasoner [84, 106, 188]. Open challenges are selectivity under cost limits and robustness to fabricated inputs (§9.2).
- **Reasoning and planning** externalize the decision trajectory in auditable natural language [141, 163]. Open challenges are building risk limits into the plan rather than post-hoc filters, and bounding deliberation cost.

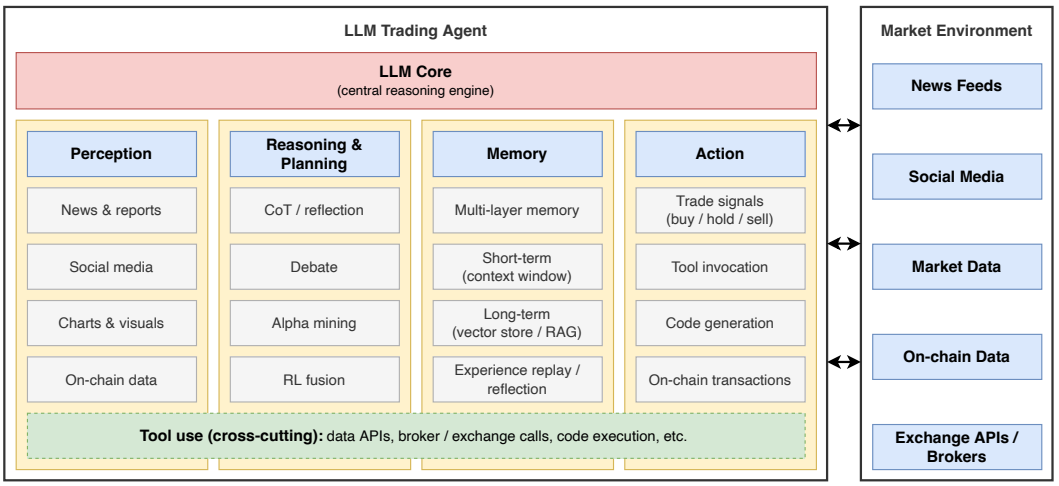


Fig. 2. Architecture of an LLM trading agent. The LLM core coordinates perception, reasoning and planning, memory, and action modules through a cross-cutting tool-use layer that connects to the market environment.

- **Action** trades expressiveness for safety: a buy/hold/sell vocabulary limits strategies, while code- and tool-based spaces enlarge the surface for errors and manipulation [2, 162]. An emerging remedy is a deterministic execution layer that bounds what the LLM can emit [109, 140].
- **Memory** moves to retrievable stores because effective context is shorter than nominal [100]; trading systems add layered hierarchies and verbal reinforcement (feedback stored as text) [182, 183]. The trading-specific challenge is *regime robustness*: keeping long-horizon patterns while discounting stale experience.
- **Tool use** spans all four modules and mirrors the action trade-off: fixed-schema function calling is easy to validate [125, 132], whereas code-as-action is Turing-complete [162].

The literature populates these capabilities unevenly across markets (Figure 3). Equities hold 32 of 47 systems and all nine memory designs. Crypto is perception-heavy (8 of 10 systems), though crypto-only systems propose no action or memory design; futures and options are action-heavy (4 of 6) with no memory design; and forex is nearly empty [69, 91]. By functional paradigm, Traders dominate crypto (9 of 10), Alpha Miners appear only in equities and futures, and bounded components dominate futures and prediction markets.

Takeaway. Each module's open problem stems from a market constraint. For action and tool use, the surveyed systems respond by bounding the LLM at its interfaces; how tight the bounds can be before language reasoning stops paying off is unresolved.

4.2 Functional Paradigms: Trader, Alpha Miner, and Bounded Component

Published systems cluster into two end-to-end *functional paradigms* [29] plus a third in which the LLM is only a bounded component of a classical system (Table 2). Traders are 22 of the 47 systems in Figure 3 and the main target of independent re-evaluation.

- **Agent as a Trader.** The agent emits trading decisions and owns the full loop, either alone (FinAgent [188], FinMem [182]) or with modules assigned to specialized roles (FinCon [183], TradingAgents [169]).
- **Agent as an Alpha Miner.** The agent produces alpha factors or signal definitions for a downstream pipeline [36, 51], from Alpha-GPT [161] to backtest-in-the-loop systems [96, 160]. The

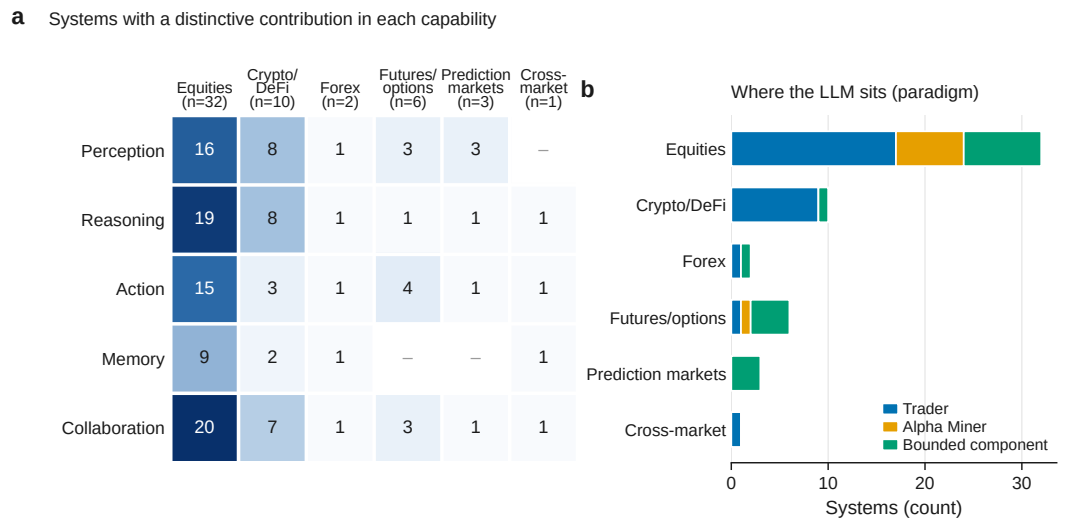


Fig. 3. Capability × market map of the 47 surveyed LLM trading systems. (a) Systems for which a capability is a distinctive contribution, per market ("–" marks an empty cell); (b) functional paradigm (§4.2) per market. Coding rules are in Supplement §S4.1.

LLM adds economic rationale, which AlphaAgent turns into a constraint: its factors hold an information coefficient (IC) near 0.02 on CSI 500 through 2024 while Alpha158, genetic-programming, and RSI factors decay to near zero [147]. Proposing code rather than trades only partly limits replay of memorized prices: in Profit Mirage's post-cutoff re-runs, Wang et al.'s QuantAgent decayed nearly as much as the Traders (§7.4) [92].

- **LLM as a bounded component.** The LLM supplies one bounded input (a signal, a view, a parsed query, or a tilt) to a classical module that owns the decision, such as an execution schedule [140], an option-strategy engine [109], a Black-Litterman optimizer [83], or a reinforcement-learning policy [10]. This functional paradigm underlies the division of labor of §8.6.

The two end-to-end paradigms diverge because their outputs differ in the cost of being wrong. A factor is cheap to re-score on history, suiting writer-critic iteration [160, 161]; a trade has immediate monetary consequence, so Traders justify it through role-specialized debate [169] and reflection over trade histories [182]. Hybrids that feed LLM signals to reinforcement-learning policies blur the boundary [45, 172], so the practical design question is which modules to assign to LLM reasoning and which to classical infrastructure.

Takeaway. The functional paradigm decides whether LLM error reaches the order book directly or is first filtered by re-scoring and classical code. No surveyed study compares the paradigms' net performance under matched conditions (Profit Mirage runs both on the same windows only to measure post-cutoff decay [92]).

4.3 Single-Agent vs. Multi-Agent Organization

Multi-agent systems are justified by specialization, heterogeneous backbones, diversity of opinion, and checks and balances [62, 85, 169, 183], whereas single agents [106, 182] are simpler but struggle to cover all the information a trading decision needs. Six organizational patterns recur (Table 3): hierarchical coordination [91, 93, 183]; debate [169, 191]; modular pipelines [38, 94]; competitive selection [190]; mixtures of fine-tuned experts [31]; and population ensembles of independent

Table 2. Design-level contrast between the three functional paradigms.

Dimension	Agent as a Trader	Agent as an Alpha Miner	LLM as a bounded component
Output	Buy/hold/sell orders with position sizes.	Alpha factors, ranking scores, or signal definitions.	A signal, view, parsed query, or bounded tilt.
Scope	Full perceive–reason–act–memory loop in-agent.	Upstream of quant pipeline; delegates execution and risk.	One input to a classical module (optimizer, RL policy, execution schedule, rule engine) that owns the decision.
Supervision	Portfolio-level: Sharpe, drawdown, cumulative return.	Signal-level: Information Coefficient, Information Ratio.	Module-level: the host module's objective (e.g., execution price vs. TWAP, query validity).
Architecture	Role-specialized multi-agent debate; reflection over trade history.	Writer–critic inner loop; backtest-in-the-loop outer loop.	Typed or bounded interface checked or executed by deterministic code.
Failures	Execution friction; decision-time reasoning errors.	Alpha decay [116]; factor crowding; look-ahead bias; data snooping across candidates [8, 57, 164].	Gain depends on the host module [10]; look-ahead in the LLM's input.
Examples	FinAgent [188], FinMem [182], FinCon [183], TradingAgents [169], ContestTrade [190].	Alpha-GPT [161] and Alpha-GPT 2.0 [184], QuantAgent [160], Kou et al. [79], R&D-Agent-Quant [96], AlphaAgent [147].	PACE [140], OQL [109], LLM Black-Litterman [83], FinRL-DeepSeek [10], Macro Economists [47].

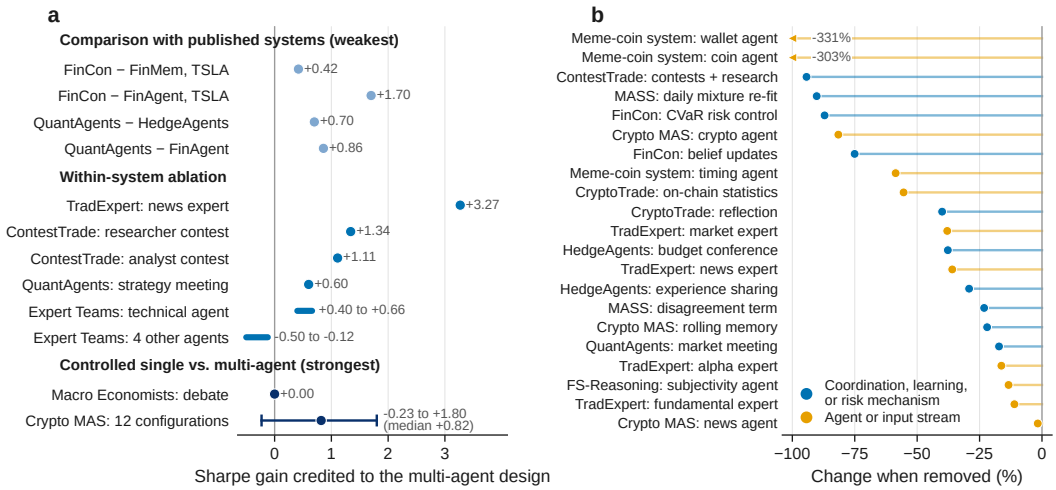


Fig. 4. Evidence for the multi-agent designs in Table 3: (a) Sharpe-ratio gain credited to the multi-agent design, by evidence grade; (b) change in headline return when one component is removed, as a percentage of the full system's value [31, 47, 55, 91, 93, 94, 110, 111, 117, 159, 183, 190]. No gap carries a standard error.

agents, as in MASS, whose rank IC rises with population size [55]. With roles fixed, hierarchy gives the best overall risk-adjusted return under all three backbones of the crypto portfolio multi-agent system (MAS), though not in every capability configuration [111].

Table 3. Representative multi-agent LLM trading systems and the strongest evidence each offers for its multi-agent design, graded as in §4.3. Roles, aggregation rules, and figures are in Supplement Table S2.

System	Market	Organization	Strongest evidence for the multi-agent design
TradingGPT [97]	US equities	Peer debate	None (design paper)
FinCon [183]	US stocks	Hierarchical	vs. prior (FinMem, FinAgent); ablation of risk control, not of agents
TradingAgents [169]	Three US stocks	Hierarchical, two debate stages	Rule-based baselines only
HedgeAgents [91]	BTC, DJ30, USD/CNY	Hierarchical, conference-based	Ablation of each conference; no single-agent row
QuantAgents [93]	NASDAQ-100; live A/HK	Hierarchical, meeting-based	vs. prior (HedgeAgents, FinAgent); ablation of each meeting
ContestTrade [190]	Chinese A-shares	Internal contest	Ablation: removing contests and deep research cuts Sharpe 3.12 to 0.07
AlphaAgents [191]	15 US tech stocks	Debate to consensus	vs. single (figure only, one run); all portfolios lose money
Macro Economists [47]	15 commodity ETFs	Persona debate	vs. single with bootstrap: no Sharpe gain over the better persona
Expert Teams [117]	TOPIX 100	Hierarchical, three levels	Leave-one-out (50 runs): four of five specialists lower Sharpe
MASS [55]	SSE 50, CSI 300, ChiNext	Population ensemble	Ablation of aggregation; IC rises with agent count (figure only)
Crypto portfolio MAS [111]	15 layer-1 coins, weekly	Hierarchical, collaborative, debate	vs. single at matched prompting (one run each): gain in 10 of 12 configurations
TradExpert [31]	DOW 30	Mixture of experts	Ablation of each expert
FS-ReasoningAgent [159]	BTC, ETH, SOL	Fact/subjectivity pipeline	Ablation (one run)
CryptoTrade [94]	BTC, ETH, SOL	Modular pipeline	Ablation of reflection and inputs
MarketSenseAI 2.0 [38]	S&P 100 and 500	Modular pipeline	Index benchmarks only
Meme-coin copy trading [110]	Solana meme coins	Independent judges	Ablation: removing wallet or coin agent turns return negative

The evidence comes in three grades (Figure 4a), and none establishes a multi-agent advantage. Prior-system comparisons (*vs. prior*) re-run others' published systems, confounding organization with data, LLM version, prompting, and window; *ablations* remove one component within the same system; and matched single-agent controls (*vs. single*) give a single agent the same inputs, LLM, and window. Over short windows without paired tests (§7.5), the gaps are neither established nor refuted. Prior-system gains are large and ablations credit the ablated mechanism, but the three matched single-agent controls disagree: AlphaAgents' multi-agent portfolio beats its single agents under risk-neutral prompts, though under risk-averse prompts every variant trails the benchmark [191]; Geng et al.'s Macro Economists debate agent adds no Sharpe over the better single persona [47]; and the crypto portfolio MAS's headline gain is the best of twelve single-run configurations, while two of its three chain-of-thought architectures trail the matched single agent [111].

One reading reconciles them: multiple agents help when they change how information is processed, not when they multiply opinions on identical inputs. The Macro Economists' personas share inputs, so debate can only average two priors, whereas the crypto portfolio MAS's agents split modalities. General-domain results agree: in a common harness no debate protocol reliably beats self-consistency [143], and majority voting over independent samples outranks debate [87]. In Agent Market Arena's two months of live paper trading [129], architecture rather than backbone drove outcome variation, yet the debate hierarchy was the most backbone-sensitive (−38.7% to +21.9% on one stock).

Gains trace to structure, input design, and risk mechanisms, and the best choice depends on the backbone. ContestTrade's gains largely vanish without its selection contests [190]. In

Miyazaki et al.'s Expert Teams, the cleanest agent-level ablation (fifty post-cutoff runs per configuration on TOPIX 100), analysts given pre-computed indicators and explicit criteria beat analysts given raw data, and removing any of four non-technical agents *raises* the Sharpe ratio, by up to 0.5 [117]; in the crypto portfolio MAS, pre-computed indicators suit GPT-4o, whereas GPT-5 favors chain-of-thought and Claude retrieval [111]. Removing FinCon's conditional value-at-risk control, HedgeAgents' extreme-market conference, or QuantAgents' risk-alert meeting causes some of the largest ablation losses [91, 93, 183].

Costs and error propagation are under-measured: costs scale with universe, frequency, and debate rounds (§8.5), and debate can converge on a confidently wrong consensus [33]. Nguyen and Pham price coordination at roughly \$0.50–2.00 per daily decision for seven agents and argue from an uncalibrated breakeven test that it pays only when the half-spread is below half the expected execution-price improvement, so mainly liquid instruments qualify [118]. Outside finance, 1,642 annotated multi-agent traces attribute failures to system design (44%), inter-agent misalignment (32%), and missing verification (24%), and an objective-level verification step raised task success by 15.6% [23]. No surveyed trading paper analyzes convergence or disagreement handling, and AlphaAgents' risk-seeking and risk-neutral agents behaved indistinguishably [191].

Takeaway. Multi-agent organization is an engineering pattern for partitioning information and inserting checks, not yet shown to beat a single agent given the same inputs, backbone, and window. The open question is which input partition and which check buy the most risk-adjusted return per token, which needs multi-seed, matched-input ablations on independent benchmarks.

5 Key Techniques for LLM Trading Agents

We organize this section by technique rather than module, because most techniques serve several capabilities of §4; collaboration is treated in §4.3.

5.1 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) counters hallucination and stale training data with external knowledge [84], but its evidence is strong for factual question answering and weak for trading. Retrieval quality bounds the gains: GPT-4-Turbo fails most FinanceBench questions from a shared vector store yet answers most once the relevant filing is in context [68], and hallucination detectors lose much of their discriminative power once noisy triplets enter retrieval [80], so retrieval relocates the reliability problem rather than removing it (§9.1, §9.2). Trading systems bundle retrieval with fine-tuning or multi-agent design [90], and only FactFin [92] isolates retrieval's contribution to returns (8.8% to 13.4% on AAPL). Prices are better served by tool calls, which answered stock-price queries with 100% accuracy [71].

5.2 Prompting and Reasoning Strategies

General reasoning capability does not predict trading performance: structure and input design matter alongside the backbone, and the best structure depends on the backbone. Domain blueprints help weak models: FinCoT lifts Qwen3-8B-Base from 63.2% to 80.5% on CFA-style questions, but the gain vanishes or reverses for models with strong built-in financial reasoning [120]. In direct cryptocurrency trading, stronger LLMs sometimes underperform weaker ones by over-weighting facts relative to sentiment, until FS-ReasoningAgent's fact–subjectivity split largely restores the expected ordering [159]. Nor do reasoning-tuned variants reliably beat their instruction-tuned counterparts on return or drawdown [25]; LMArena rank does not predict live portfolio performance [181], and a 135M-parameter trading-trained model beats GPT-4 agents (§5.3). Reasoning distillation reliably lifts financial reasoning *benchmarks* [103, 130, 192], though Fin-R1's ablation shows that RL without a supervised stage yields incoherent reasoning even as accuracy rises, and

general reasoning models regress on financial tasks relative to their instruction-tuned bases [103]. General leaderboards are therefore a poor guide to backbone choice: only Trading-R1 [168] backtests a reasoning-trained trading decision, and no surveyed study varies scale with structure held fixed.

5.3 Fine-Tuning and Domain Adaptation

Fine-tuning pays at the perception and action ends of the pipeline and fails out of distribution. At the perception end, LLMs fine-tuned on newsflow beat conventional sentiment scores as long-short signals [54], and FinLoRA adapters on an 8B model lift an aggregate financial-task score from 37 to 75 (zero-shot GPT-4o: 63) [157]. At the action end, FLAG-Trader trains SmolLM2-135M as a reinforcement-learning (PPO) policy on Sharpe-ratio change and beats GPT-4 and GPT-4o agents for four of six assets [172]. Out of distribution, Palmyra-Fin-70B leads InvestorBench on ETFs yet loses 20.8% on Bitcoin [86], two finance-tuned models fall to 10–12% accuracy on FinCall-Surprise by over-predicting negative labels and breaking the answer format [142], and tuning on historical news raises in-distribution accuracy by about twenty points while cutting unseen-data accuracy by 18–22% relative [92]. An agent that operates across regimes therefore cannot treat in-distribution gains as skill.

5.4 Tool Use and External Integration

Tools help most when they move computation into a deterministic, verifiable engine. The Option Query Language (OQL) [109] *narrows* the action space to typed queries that an engine validates against the option chain (§6.3), whereas LLM-built stochastic-differential-equation models *widen* it, lifting an LLM trader's mean Sharpe ratio from 0.88 to 1.40 across four stocks (as reported; level with buy-and-hold) while hurting weaker models (Table S5) [34]. Typed interfaces thus lower the skill the model needs, whereas open-ended tool construction pays only for strong models; in both cases numerical errors survive delegation (§9.1).

5.5 Memory Architectures

Memory's trading evidence comes from ablating the *update loop*, not storage. Agents use layered memory (FinMem [182]), reflection on prior outcomes (CryptoTrade [94]), or verbal reinforcement (FinCon [183]). Removing reflection cuts CryptoTrade's bull-market return from 28.5% to 17.1% (Table S2), and FinCon's belief updates add 25–85 percentage points of cumulative return. Layered tiers amount to paging policies [122] with hand-set decay constants, yet no surveyed trading agent tests whether they beat a flat store, and no open benchmark evaluates memory modules directly [189]. The main open problem is learning when to forget: pruning stale market beliefs while keeping risk principles [65].

5.6 Perception Techniques: Sentiment, Multimodal, and Alternative Data

Every ablated perception modality helps, and its mechanism predicts its durability. GPT-4 headline sentiment earns a pre-cost long-short Sharpe ratio of 2.97, falling from 6.54 in late 2021 to 1.22 by early 2024 as LLM adoption spread [106], and on close to a million articles OPT sentiment beats FinBERT and the Loughran–McDonald dictionary after costs (long-short Sharpe 3.05 vs. 2.07 and 1.23) [76] (head-to-head evidence in §8.1). Adding charts and reflection raises GPT-4o's return in a virtual market from 33.7% to 47.7% (charts alone: 35.8%); the authors attribute the chart gain to weak reading of numerical text [112]. Dropping on-chain statistics costs CryptoTrade more return than dropping news (28.5% falls to 12.7% versus 19.7%) [94]. Sentiment alpha thus decays with adoption, chart gains compensate for a numerical-reading deficiency that better tooling should erode, and on-chain data exploit crypto's transparency, but their persistence is unmeasured. No

surveyed system models each source's staleness, and perception remains the least benchmarked module.

6 Applications Across Financial Markets

Markets differ in price drivers, action space, microstructure, and available test data; within each market we group systems by how they respond to these constraints. Fixed income and market making are omitted because no surveyed LLM-agent study trades bonds or makes markets; market makers appear only as simulated personas [105].

6.1 Equity Markets

Equities pair the richest text with the simplest action space and are the default testbed [29] (Table S3). Traders differ mainly in where they adapt (memory [182], organization [169, 183], or weights [172]), and three regularities follow. First, richer inputs and rationales beat their own ablations without lifting direction calls far above chance: training SEP to explain lifts accuracy over a binary-only variant (40.8% to 44.1%), yet the full model reaches only 51.4% (Matthews correlation 0.03) [77], and extra specialist analysts can lower Sharpe (§4.3). Second, the locus of adaptation sets cost: TradingAgents makes 11 LLM calls and 20+ tool calls per decision [169], whereas weight-trained designs pay once in training and infer with 7B or 135M models [77, 172]. Third, rigor tracks portfolio scope: no single-stock system prices transaction costs, and their tests span 3–8 months (Table S4), whereas portfolio systems such as MarketSenseAI 2.0 [38] more often state costs and run longer. Equities thus supply nearly all architectural variety but mostly short-window evidence. The one leakage-controlled, fifty-run study, Expert Teams [117], trails its index on net Sharpe while every blend with the index beats both, so its demonstrated value may be diversification rather than alpha.

6.2 Cryptocurrency and DeFi

Crypto adds 24/7 trading, a public ledger as data, and direct protocol actions. Exchange agents resemble equity agents: CryptoTrade beats buy-and-hold in only four of nine asset-regime cells [94], and the strongest post-cutoff result is fragile. Over 2025, after all backbones' cutoffs, the best configuration of the crypto portfolio MAS returns 133.5% net of costs against −3.4% for holding Bitcoin, but with a deeper drawdown (39.3% vs. 29.7%) and one run per configuration [111]. On-chain perception is further along than execution: the TIM framework infers DeFi transaction intent better than the best supervised classifier [115], whereas Alqithami advises high-value systems to stop short of autonomous on-chain signing [2]. Deployed practice runs ahead of this caution: ElizaOS lets agents hold wallets and swap tokens [156], yet an instruction planted in its shared memory can trigger a 50x-leverage copy-trade [126] (§9.2). Maximal extractable value (MEV) is an exposure, not a strategy: we find no LLM-driven extraction [49], and since replay front-running needs no insight into the victim's logic [131], reasoning cannot protect agents that submit swaps. In crypto, LLMs thus add the clearest value in reading the ledger, while on-chain autonomy lags by design; where deployed frameworks bypass that design, shared memory becomes the attack surface.

6.3 Emerging Entry Points: Forex and Derivatives

Apart from HedgeAgents' USD/CNY sleeve [91] and QuantHarness's cost-free direction calls on futures [171], no end-to-end LLM agent trades forex or derivatives in the studies we found; LLMs enter at the signal, factor, allocation, or order level. We conjecture that forex agents are scarce because high-frequency institutions dominate at latencies far below LLM inference times and major pairs' lower volatility limits sentiment-driven alpha. In forex, GPT-4o labels of 174,820 G10 macroeconomic releases yield a long-short portfolio that beats carry and value on Sharpe, though

without an agent loop or costs [69] (Table S5). In futures, forty GPT-generated factors earn significant alpha on Chinese futures [27], but against non-LLM rules on identical inputs the edge is thin (§8.1): QuantHarness's agents trail the better of linear regression and XGBoost on Sharpe for every instrument with baselines [171]. In options, OQL [109] translates rather than trades, lifting win rate to 60.9% from 52.8% for Text-to-SQL. Every entry point thus leaves pricing, Greeks, and margin arithmetic to deterministic code (§5.4), and no derivatives system shows a robust, cost-adjusted gain over a non-LLM rule where one is tested.

6.4 Prediction Markets

Prediction markets reward LLMs' text synthesis, yet agents lose when forecasts must beat prices. Forecasting approached superforecaster level within about eighteen months. On ForecastBench's unresolved questions, frontier LLMs in July 2024 matched an inexperienced crowd (Brier 0.122 vs. 0.121) but significantly trailed superforecasters (0.096) [72]; by late 2025 the AIA Forecaster statistically matched superforecasters yet trailed the market price on 1,610 liquid-market questions, and only its ensemble with the market beat the market [3] (Table S5). Trading profit is harder: in Prediction Arena all six frontier models lost real money on curated Kalshi markets [187], and PolyBench's two profitable models lose their edge at larger order sizes [26]. Positive results exploit structure: an agent that recovers logical relations among Kalshi contracts nets 14.1% in two months of fee-aware arbitrage, though five of 46 trades supply 76% of the profit [22]. Because the price already aggregates what an LLM reads, agents profit only from information the market lacks or from unpriced structure.

6.5 Summary

Maturity falls from equities through crypto to derivatives, prediction markets, and forex (32, 10, 6, 3, and 2 of the 47 systems in Figure 3; multi-market systems count in each), and one rule explains the ordering: LLM agents go furthest where text moves prices and actions are simple, and elsewhere they only interpret (releases, news, intents, contract relations) or propose (factors, views, queries) while deterministic code executes [22, 83, 91, 109]. As in prediction markets, credible profits come from unpriced structure (contract relations, manipulation patterns) rather than better prediction [3, 22, 110]. Cross-market work confirms the rule: HedgeAgents [91] allocates across Bitcoin, Dow Jones stocks, and USD/CNY around a classical allocator, with the LLM supplying views and hedging triggers (Table S3). Adding asset classes is easy; robustness is not, since an agent beating a US index can trail the SSE 50 and lose in crypto [37].

7 Evaluation Frameworks and Empirical Analysis

What a reported result can claim is bounded by the instruments and protocol behind it, so we examine metrics, environments, and benchmarks first, and then the results and the protocol pitfalls that explain why author-run and independent results diverge.

7.1 Evaluation Metrics

The standard metrics measure return and risk but omit what decides whether a return is deployable (Table S1 defines the terms). Papers converge on cumulative and annualized return, the Sharpe ratio, and maximum drawdown; Alpha Miners add the information coefficient and its variants [160, 190]. Among the surveyed studies, turnover appears in only two [79, 88], capacity in one [106], and factor exposure in three [38, 69, 88], while system metrics (token cost, latency) rarely appear beside returns. Without turnover, factor exposure, and cost beside returns, most headline Sharpe ratios in this literature cannot be distinguished from a cost-free market or momentum bet.

7.2 Backtesting Frameworks and Simulation Environments

Environments trade execution fidelity against accessibility. FinRL [101, 102] and FinRobot [176] model no execution, ABIDES [18] simulates exchanges for traditional agents, and StockSim [123] pairs limit-order-book execution with LLM agents. Shared-market simulators [105, 112, 178, 185] capture the price impact that single-agent backtests ignore: TwinMarket's persona-driven agents reproduce fat tails and volatility clustering better than classical heterogeneous-agent models [178]. Macro-level realism, however, does not imply human-like micro behavior. LLM populations in experimental markets price near fundamentals where human subjects produce bubbles [60], and persona-conditioned agents reproduce loss aversion but never migrate toward fundamentals as prices diverge from value [98].

Takeaway. No environment yet combines execution friction, multi-agent price impact, and multimodal inputs. LLM simulators reproduce what prompts and personas encode rather than human behavior, so without per-model behavioral validation they cannot replace human-subject experiments in market design.

7.3 Major Benchmarks and Datasets

Benchmarks form four groups, the first three progressively closer to trading, and success in one does not carry to the next. **Knowledge and reasoning benchmarks** such as FinBen [170] involve no trading and can mislead: on FinCall-Surprise's skewed labels, most models exceed 70% accuracy yet score below 0.55 in macro-F1 [142]. **Agentic research benchmarks** locate the bottleneck in retrieval: on Finance Agent Benchmark the best model reaches 46.8% accuracy at $3.79 per question [11], and models without search score zero on FinSearchComp's time-sensitive questions [63]. **Trading-specific benchmarks** evaluate decisions under market conditions: StockBench [25] and InvestorBench [86] backtest, LiveTradeBench [181] and AI-Trader [37] trade on live data, and the Agent Market Arena [129] places published architectures under one live paper-trading protocol. **Domain-specific benchmarks** target segments such as on-chain tasks [52] and prediction markets, from scoring unresolved questions [72] to replaying order books [26] and trading real capital [187].

Takeaway. Neither QA performance nor forecasting accuracy reliably predicts trading profit [26, 187], and trading benchmarks cover few asset classes over short periods, so an integrated, cost-aware evaluation of the full perception–reasoning–action loop against market prices is still missing.

7.4 Empirical Results Summary

Author-run comparisons in Table S3 favor multi-agent and multimodal designs, whereas independent evaluations find thin, regime-dependent edges. FinCon [183] and TradingAgents [169] beat their baselines on return and Sharpe ratio, and MarketSenseAI 2.0 [38] returns 125.9% on the S&P 100. By contrast, 8 of StockBench's 13 generic agents beat buy-and-hold, mainly through lower drawdowns, yet all trail it in the downturn window [25], and none of AI-Trader's live agents profits in crypto [37]. Re-running published systems gives the sharpest contrast: FINSABER, Profit Mirage, and Alpha Illusion shrink or erase the author-run margins [88, 92, 180]. In Profit Mirage's matched windows on either side of the training cutoff, five agents, including Wang et al.'s QuantAgent [160] (the only Alpha Miner re-run, about 51%), lose 51–62% of their Sharpe ratios [92].

Read beside its baseline (Figure 5), each headline conceals two problems. First, the largest margins come in windows where the passive benchmark lost money or barely gained (FinMem's 61.8% on TSLA against a −18.6% buy-and-hold), so they largely reflect loss avoidance; in rising markets, MarketSenseAI 2.0 draws down deeper than the index and Expert Teams trails the

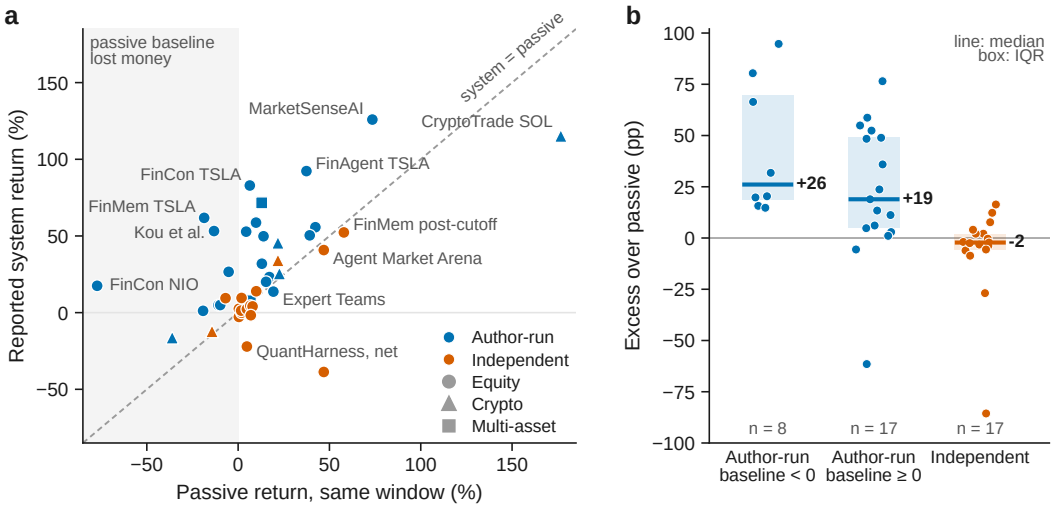


Fig. 5. Headline returns against the passive baseline each paper reports over the same window (Table S3). (a) Points above the dashed line beat their baseline, and shading marks windows in which the baseline lost money; (b) excess return by evaluator and baseline sign, with medians that are descriptive only.

TOPIX 100 [38, 117, 182]. Second, costs are often undisclosed: only 8 of the 19 author-run systems in Table S3 state a transaction-cost value, all from 2025–2026 (Supplement §S11).

Takeaway. Where published agents have been independently re-run over longer, post-cutoff, or cost-charged windows, their margins shrink or vanish. The finding is narrow. The re-runs cover a handful of early, mostly single-stock systems. Two of the three re-evaluations are preprints by authors proposing competing designs, so independent does not mean disinterested, and re-evaluators target the most prominent results, inviting regression to the mean. Because many original margins lie within the sampling error of their short windows (§7.5), the re-runs show that margins are not reproduced, not that a real edge decayed. Neither author-run nor independent results yet establish that LLM agents beat passive strategies, or that they cannot.

7.5 Evaluation Protocols and Their Pitfalls

We audit the protocols behind this divergence on six dimensions (Table S4). Five pitfalls recur; correcting the three that re-evaluations have tested lowers headline Sharpe ratios (Figure 6), and the evaluation mode bounds the execution realism of any result (Figure 7).

Short windows and small universes. Eight of the seventeen dated author-run evaluations in Table S4 last eight months or less, several in rallies, and single-stock agents trade three to eight news-rich large caps, which reveals nothing about breadth or capacity. Lengthening the window erases the edge: over 2004–2024 with commissions on survivorship-free S&P 500 universes, FinMem's 2.68 Sharpe on TSLA becomes 0.93 or 0.40 depending on the backbone, and buy-and-hold beats both FinMem and FinAgent on every composite universe [88]. TradingAgents' authors themselves flag their 5.6–8.2 Sharpe ratios as artifacts of a period with few pullbacks [169].

Unmodeled costs. Costs can reverse conclusions. Charging full frictions over one year on five tickers cuts TradingAgents' Sharpe from 0.43 to 0.22, below equal-weight buy-and-hold, and 35 of the 40 system-by-friction cells in that audit of five systems are unmodeled [180]. Every reported gross Sharpe is therefore an upper bound on deployable net Sharpe; ML return predictability in

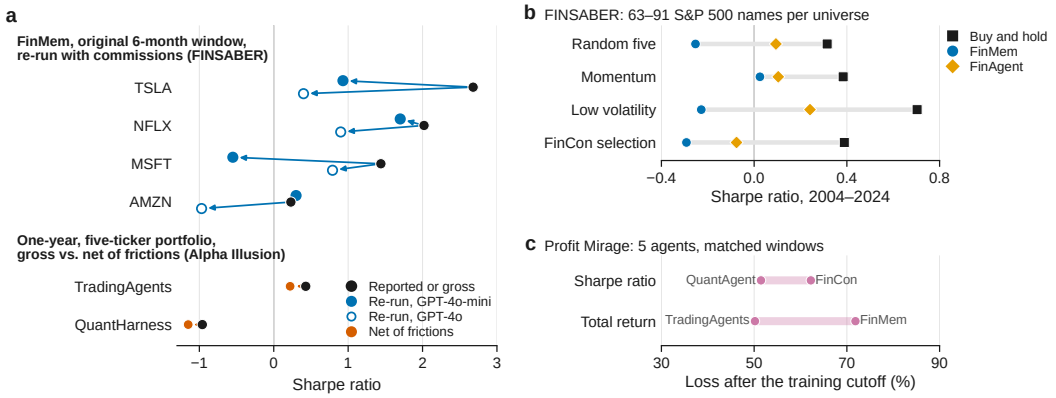


Fig. 6. Headline Sharpe ratios under independent re-evaluation. (a) As reported versus re-run with costs [88, 180]; (b) FINSABER's 2004–2024 composites under four stock-selection rules [88]; (c) the range of Profit Mirage's post-cutoff decay across five agents [92].

traditional finance likewise attenuates under realistic costs [6]. Even stated rates are flat charges that ignore how impact grows with capital.

Contamination and look-ahead. An LLM pre-trained over the test window has already read the news, and possibly the prices, it trades on. Profit Mirage traces its post-cutoff decay to recall: backbones recall dated prices, trends, and event reactions with 85–93% accuracy [92]. Yet only 26.8% of 164 main-conference LLM-finance papers discuss look-ahead bias [78]. Instructions to ignore the future and anonymization both fail across models and tasks [89, 107, 135]: GPT-4o told to ignore post-2010 data forecasts GDP direction as well after that date as before (98.0% vs. 97.6%, against 40% past its real cutoff) [107], and stronger models re-identify 70% of name-censored earnings calls [135]. Only post-cutoff evaluation, live evaluation, and chronologically consistent backbones [59] control leakage by construction (mitigations in Supplement §S7.5).

Single runs and self-selected baselines. Decoding is stochastic, yet only 5 of the 28 studies in Table S4 report repeated runs, and two runs of one FinRL-DeepSeek configuration reorder the agents [10]. Baselines are self-selected, and raw buy-and-hold is the wrong benchmark for agents that vary their exposure (§8.4); the baseline should match the agent's average beta. The fifty-run rank tests of Expert Teams [117] and the block bootstrap of Macro Economists [47] are the first significance tests on return differences among the surveyed studies.

Statistical power and selection. Even a correctly run backtest over these windows cannot separate most agents. Under iid returns, the standard error of an annualized Sharpe ratio from T years sampled q times a year is about $\sqrt{(1 + \mathrm{SR}^2/2q)/T}$ [104], roughly 1.4 for six months of daily returns; FinMem's 2.68 on TSLA over six months thus carries a 95% interval of roughly −0.1 to 5.5, which already contains FINSABER's re-run values, and serial correlation and fat tails widen it further [9, 104]. A margin over a baseline, being a difference of two such estimates, has a standard error of about 2.0 over six months for uncorrelated series; by our calculation, 9 of the 14 author-run Sharpe gaps over a passive baseline in Table S3 lie within two such standard errors. The same noise bounds the corrective evidence. Published equity anomalies lose 26% of their returns out of sample, an upper bound on data mining, and 58% after publication [116]. Profit Mirage's 51–62% decay exceeds the former and matches the latter, but it is one correlated draw of five agents on one pair of six-month windows, so decay from post-publication arbitrage cannot be separated from the loss of memorized information. Selection compounds the noise, because the expected

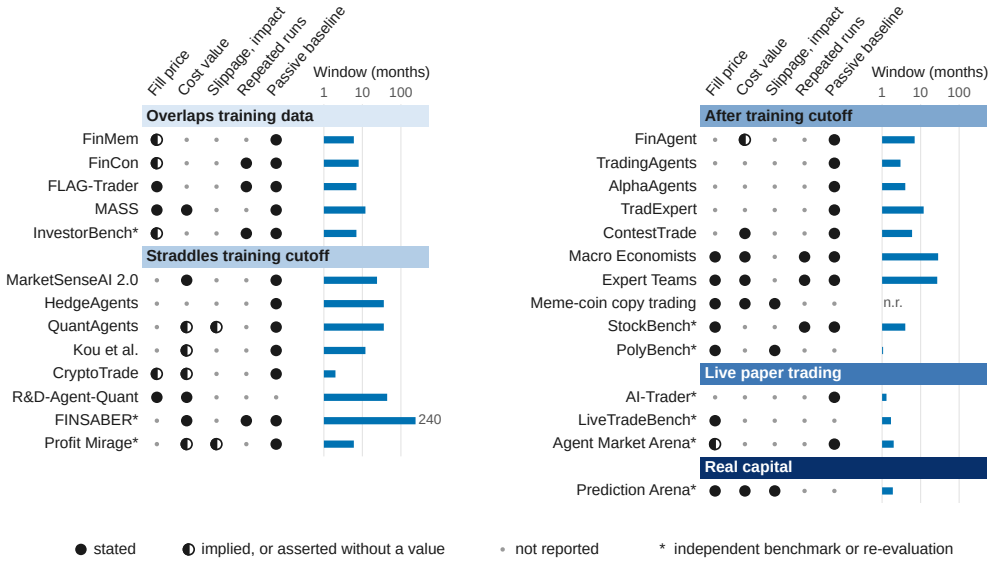


Fig. 7. Execution disclosure by evaluation mode for the systems of Table S4, ordered by execution realism. Marks record whether fill price, cost, slippage, repeated runs or tests, and a passive baseline are stated (filled), inferred or asserted without a value (half), or absent (empty); bars give test windows on a log scale.

maximum Sharpe ratio of N zero-skill configurations rises with N [9]. For example, a seasonal rule optimized on a pure random walk reaches an in-sample Sharpe ratio of 1.27 that a single-trial test calls significant, yet its probability of backtest overfitting (PBO) is 55% [8]. The deflated Sharpe ratio and PBO correct for both effects only given the number and returns of all configurations tried [8, 9], which no system in Table S4 reports.

Evaluation mode and fill realism. Evaluation modes form a ladder (Figure 7) from pre-cutoff backtests through post-cutoff backtests and live paper trading to real capital, which only 1 of the 28 studies in Table S4 reaches. In Prediction Arena, orders were rejected for lack of counterparties, and spread and fee drag were large enough that skill cannot be separated from execution [187]. Paper trading removes look-ahead but not execution: it fills at closing prices or scoring signals without sizing or costs [129, 181]. Live paper trading nonetheless reorders agents: risk control rather than reasoning quality separates robust from fragile agents in AI-Trader, and general-capability rank does not predict live portfolio quality in LiveTradeBench [37, 181]. Below the live rungs, only 6 of the 18 author-run systems state a fill rule, and none reports fill rates or partial fills; in PolyBench's order-book replay, both profitable models' alpha collapses as order size grows from \$10 toward \$1,000 [26].

Corrections and audits agree. Each correction in Figure 6 leaves at least one corrected agent trailing buy-and-hold, though their combined effect is unknown, and Xia et al.'s audit of 185 studies [167] finds reporting improving but incomplete, so the proportions in this section describe the most-cited early systems rather than current practice. Published checklists [45, 78, 167, 180] converge on a minimum: a post-cutoff or live window spanning a downturn, the full universe with capacity analysis, explicit cost and slippage, run counts and dispersion, the number of configurations tried, and exposure-matched passive plus tuned active baselines.

Takeaway. The headline returns in Figure 5 are existence proofs under favorable conditions, not estimates of expected performance; even live returns are upper bounds conditional on full fills,

Table 4. LLM trading agents vs. traditional quantitative trading across five dimensions, with the figures reported in the surveyed studies. Costs are LLM API spend as stated by each paper.

Dimension	Traditional Quant	LLM Trading Agents
Data	Structured price, volume, order-book, and factor data; text enters through hand-engineered sentiment features that do not significantly predict next-day returns ($t = 1.87$) where LLM scores do (Loughran–McDonald [108] long-short Sharpe 1.23 vs. 3.05 for OPT) [76].	Text, numbers, and chart images without task-specific features [188]; headline sentiment alone yields a pre-cost Sharpe of 2.97 over 2021–2024, falling from 6.54 (2021Q4) to 1.22 (2024) [106]. Weak at exact arithmetic: agents miscalculate share quantities and read absolute levels over percentage changes [25, 112].
Interpret.	Factor strategies are transparent by construction; DRL policies (A2C, PPO, SAC) are opaque.	Natural-language theses [168], but 69–82% of decisions survive counterfactual changes to events, prices, and fundamentals, so stated rationales often do not drive the decision [92].
Adapt.	Regime-specific calibration; buy-and-hold Sharpe 0.61 in bull years, −0.28 in bear years over 2004–2024 [88].	In-context adaptation and reflection [94, 159], yet FinAgent earns 0.12 in bull years and −0.38 in bear years on the same sample [88]; model rankings reorder between downturn and upturn windows [25]. Robustness in extreme markets comes from hard-coded conferences and thresholds [91, 93].
Latency and cost	Order-book time scales, simulated at nanosecond granularity [18]; execution schedulers such as TWAP and Almgren–Chriss run at minute granularity [140]; no per-call inference fee.	Seconds to minutes per decision: 4.7 s per stock on one GPU [31]; 11 LLM and 20+ tool calls per decision [169]; one decision per 15–45 minutes in live prediction-market agents [187]. API spend: $15 (three asset classes) and $180 (NASDAQ-100) over three-year daily backtests [91, 93]; $0.68–2.27 per trading day for 50–300 stocks [55]; $198 vs. $32 for a twenty-year composite re-run of FinAgent vs. FinMem [88]; about $30 for 1,680 execution orders [140].
Scale	Linear to thousands of instruments; FINSABER re-runs agents on historical S&P 500 universes of 63–91 names including delisted stocks [88].	The most-cited single-stock agents (FinMem, FinCon, FinAgent, TradingAgents) trade three to eight names (Table S4); returns fall and volatility rises from 10 to 30 stocks, though 5-stock portfolios fare worst [25]; FinCon's authors note hallucination becomes likelier as inputs lengthen [183]; a 512-agent ensemble exceeds the context window on CSI 300 without per-agent stock subsets [55]. Exceptions cover the S&P 500 with monthly signals [38].

and no live window yet spans a market cycle. The infrastructure exists (§7.3); what is missing is a norm that labels claims below this minimum as exploratory.

8 LLM Agents vs. Traditional Quantitative Trading

This section compares LLM agents with traditional quantitative trading on reported evidence rather than claimed capability. Table 4 contrasts the two approaches on five engineering trade-offs; because it draws mainly on LLM agent papers, we first consolidate the head-to-head record and then judge each dimension (§8.2–§8.5).

8.1 Head-to-Head Evidence

Signal-level evidence is the strongest; author-run agent leads are common but rarely net of costs. LLM text signals beat dictionary [108] and FinBERT [5, 64] sentiment on return predictability and Sharpe ratio in most cells across groups, data sets, and model families (Table 5) [54, 76, 106], though the edge decays with adoption (§5.6). Of the 17 agent-level studies, 11 lead their strongest traditional baseline, 4 are mixed, and 2 trail. Only four leads hold at a stated cost rate (three of them by Alpha Miners or bounded components, one not significant), and only three studies report a significance test (Supplement §S8.1). Where agents lose, the winner is often a simple rule or passive holding [94, 172]: QuantHarness, for example, trails linear regression or XGBoost on Sharpe for all seven instruments [171]. Wins over deep reinforcement learning (DRL) baselines are confounded, as several A2C and PPO baselines see only prices and use default settings.

Independent evaluation shrinks the end-to-end edge. Re-runs of published systems shrink or erase author-run margins over passive baselines [88, 92, 180] (§7.4). Contamination-controlled

Table 5. Head-to-head outcomes of LLM-based approaches against the strongest non-LLM baseline each paper reports (passive included), on its headline metric (risk-adjusted if reported, else return; mean over backbones if the paper designates none). Codes use point estimates over cells (assets, universes, settings): *ahead* wins every cell, *behind* none, *mixed (k/n)* wins k of n; n.s.: the paper's own test finds the margin insignificant at 5%. Cost flags (*net*: the code holds at the stated rate; *gross*; *exec.*: the metric is itself an execution cost; *unspec.*) do not change the code; n.c.: not coded; n.r.: not reported. Functional paradigms (A, B) follow Figure 3b; rows are author-run unless marked ext. (external scoring). Windows, baselines, and figures: Supplement Table S5.

Study	Market	Traditional baselines	Code	Costs	Note
Signal-level studies (LLM as perception or predictor)					
Lopez-Lira and Tang [106]	US stocks	FinBERT [5], market	Ahead (1/1)	net	10 bps; unprofitable at 20 bps; edge decays
Kirtac and Germano [76]	US stocks	Dictionary [108], FinBERT [64], market	Ahead (1/1)	net	10 bps
Guo and Hauptmann [54]	NA, EU, EM equities	FinBERT [5], FinVader, universe	Mixed (4/6)	gross	Loses NA and EM long-only to FinBERT
Alpha-GPT [161]	Quant competitions	Human quant teams	Behind (0/2)	n.r.	ext.; top ten but below each winner
Izadyar [69]	G10 currencies	Carry, value, momentum	Ahead (1/1)	gross	No surprise-rule baseline
Makri et al. [114]	ETH	PatchTST, LSTM, MLP	Ahead (2/2)	–	Forecast only; no trading test
AIA Forecaster [3]	Forecasting questions	Superforecasters, market prices	Mixed (2/4), n.s.	–	Forecast only; ext.; trails superforecasters on FB-Market, the market on liquid markets
Agent-level studies (Trader; A: Alpha Miner; B: bounded component)					
FinMem [182]	5 US stocks	Buy-and-hold; DRL	Ahead (5/5)	gross	Risk setting chosen on test window
FinCon [183]	8 US stocks	DRL; Markowitz	Ahead (10/10)	gross	
FinAgent [188]	5 US stocks, ETH	Rule; ML; DRL	Mixed (3/6)	gross	Ahead on return in 6 of 6
TradExpert [31]	DOW 30	ML; DRL; index	Ahead (1/1)	gross	
CryptoTrade [94]	BTC, ETH, SOL	Rules; deep forecasters	Mixed (1/9)	unspec.	Tops buy-and-hold in 4 of 9 cells
R&D-Agent-Quant (A) [96]	CSI 300/500, NASDAQ 100	Factor libraries; 16 models	Ahead (3/3)	net	No passive benchmark
MASS (A) [55]	Chinese indices	Rule; index; ML; DL	Ahead (3/3)	net	Rank IC agrees
Kou et al. (A) [79]	SSE 50	ML; DRL; index	Ahead (1/1)	unspec.	Turnover-capped; half-years vs. index not coded
ContestTrade [190]	A-shares	Rules; ML; DRL	Ahead (1/1)	net	Larger drawdown than a rule
HedgeAgents [91]	BTC, DJ30, FX	DRL; momentum; buy-and-hold	Ahead (1/1)	gross	2024 hold-out not coded
QuantAgents [93]	NASDAQ-100	DRL; mean-variance; index	Ahead (1/1)	gross	Live track not coded
FLAG-Trader [172]	5 US stocks, BTC	Buy-and-hold	Mixed (5/6)	gross	Ahead on return in 6 of 6; no active non-LLM baseline
FinRL-DeepSeek (B) [10]	NASDAQ-100	PPO; CVaR-PPO; index	Behind (0/1)	gross	Two runs reorder the agents
Macro Economists (B) [47]	Commodity ETFs	Same-input rule; passive	Ahead (1/1), n.s.	net	+0.04 Sharpe; $p \approx 0.07$–0.09 unadjusted
QuantHarness [171]	Index futures, crude, BTC	Linear regression; XGBoost	Behind (0/7)	gross	Ahead on directional accuracy in 8 of 8
Emmanoulopoulos et al. [34]	4 US stocks	Buy-and-hold	Mixed (2/4)	net	Mean over backbones; best backbone wins 4 of 4
PACE (B) [140]	SZSE execution	TWAP; Almgren–Chriss; ML	Ahead (2/2)	exec.	Execution cost; sig. vs. TWAP only; no market impact
Not coded (abstract only)					
Cheng et al. [27]	Chinese futures	IPCA factors	n.c.	n.r.	Reports significant alpha; magnitudes n.r.

benchmarks agree: generic single-agent workflows beat passive baselines by at most about two points in StockBench and lose in its downturn [25], and in AI-Trader they beat no index in A-shares and make no profit in crypto [37]. Alpha Miners are not exempt: the one re-run decayed past the backbone's cutoff like the Traders, while R&D-Agent-Quant's out-of-sample lead over factor libraries and sixteen machine-learning and deep models [96] remains a single author-run study. Because Alpha Miners score many candidates and report survivors, they automate the multiple-testing ("factor zoo") problem, under which a new factor needs a t-ratio near 3.0, not 2.0 [57, 164]; yet no Alpha Miner study we reviewed reports its candidate count with a selection correction (§7.5). Independent and author-run results agree only that LLM agents, holding less exposure, show shallower drawdowns in tested windows [25, 91]; in sustained declines they do not protect capital better than passive exposure [25, 88].

Takeaway. The LLM results not yet contradicted all feed or bound a conventional pipeline. With inputs matched, the decision layer adds little: Geng et al.'s Macro Economists beat a same-input rule by an economically small margin, significant only at the 10% level [47], and PACE beats TWAP by about one basis point (95% CI 0.15–2.12) by tilting a classical schedule [140]. Each such result yields a score, formula, or bounded tilt that a conventional backtester or matcher evaluates. Autonomous LLM traders beat their chosen baselines in most author-run tests, but only one of their six leads in Table 5 is net of costs, and they lose ground under independent re-evaluation.

8.2 Data Dependency and Information Processing

LLMs win on text and traditional systems on numbers; the text edge is well evidenced but perishable. LLM agents take text, numbers, and images directly [188], and the edge is largest when the LLM extracts features rather than decides: embeddings of full news articles fed to a conventional return regression beat past-return, dictionary, and word-embedding signals across 16 markets and 13 languages [24], although FinBERT still wins on emerging-market long-only Sharpe [54]. On numerical forecasting the ranking reverses because of text pretraining rather than the transformer architecture: models pretrained on time series beat prompted or reprogrammed text LLMs in zero-shot forecasting [4, 28], and replacing the LLM in Time-LLM [70] with one attention layer beats it in all 26 dataset-metric cases [146]. Base LLMs prompted with digit strings forecast competitively zero-shot, but the chat-aligned models that agents run on do worse [50].

8.3 Strategy Interpretability and Transparency

LLM explanations are the most legible but not demonstrably faithful, so only an inspectable strategy makes them trustworthy. Unlike opaque DRL policies, LLM agents produce structured theses stating *why* a trade is recommended [168]. Yet 69–82% of decisions survive counterfactual changes to their inputs [92], chain-of-thought explanations can rationalize answers driven by input features they never mention (only 1 of 426 explanations supporting a biased answer names the bias) [149], and an answer's dependence on its stated reasoning often falls as models grow [81]. Alpha-GPT [161, 184] sidesteps the problem: the LLM writes alpha expressions, so auditors inspect a formula, not a narrative. Whether any faithfulness test can certify an end-to-end trader remains open.

8.4 Market Adaptability and Generalization

No evidence shows LLM agents adapting better to regime change; the strongest evidence shows them adapting worse than passive exposure. LLM agents can in principle interpret novel situations and reflect [94, 159], yet CryptoTrade's reflective agent leads the best rule or forecaster in only one of nine asset-regime cells, and over twenty years FINSABER finds LLM timing agents "too conservative in bull markets and too aggressive in bear markets" [88]. Reported

Table 6. A reference architecture for the division of labor: for each LLM role, the artifact it emits, the deterministic gate it passes, the bound on its action, what to monitor, and the current evidence. Gates, bounds, and monitoring are our recommendations; controls common to every role (testing before deployment and after substantial or autonomous model changes, pre-trade limits, kill functionality, algorithm identifiers [35, 43, 128, 152]) are not repeated.

LLM role	Artifact	Validation gate before use	Bound on action	Monitoring	Evidence
Text and event signals	Score or label per document	Out-of-sample IC or long-short Sharpe net of costs on post-cutoff data, incremental to existing sentiment and price factors, with a capacity curve (§7.5)	Input to a conventional model; no order authority	Rolling net Sharpe by year [106]; return captured at the realized decision lag; drift after backbone updates	Signal-level, long-sample; costed in 2 of 3 [24, 76, 106]
Factor and strategy code (Alpha Miner)	Formula or code	Backtest that counts every candidate tried: deflated Sharpe or PBO [8, 9], a t-ratio near 3 [57]; capacity curve; human review of the formula	Factor enters an optimizer under position limits	Live versus backtest IC; retire decayed factors	Single author-run [96, 147]; the one independent re-run decayed [92]
View or tilt on an optimizer or allocator	Return views with confidence; signals to a policy	Comparison with the same optimizer without views, on post-cutoff data	Optimizer shrinks views toward the prior; position limits	Backbone-specific bias in views [83]	Single author-run, mixed [10, 47, 83]
Execution tilt	Size adjustment around a schedule	Shared-matcher test against TWAP and Almgren–Chriss on identical orders, with an impact model; then transaction-cost analysis (TCA) against arrival price in a capped live pilot	Planner shifts $\leq$30% toward its own weights; Executor sizes within $\pm$50% of each slice [140]	Slippage against the schedule; completion	Single author-run, no market impact [140]
Query or intent parsing	Typed query (e.g., OQL)	Schema and constraint check by a deterministic engine [109]	Engine selects among templated strategies	Validity rate of parsed queries	Single author-run benchmark, no costs [109]
Research synthesis for humans	Report or thesis	Human sign-off; plain description of intended function [43]	No order authority	Faithfulness checks [78]; reliance on unflagged errors [124]	Field evidence on analysts [138]

extreme-market robustness comes from explicit triggers and hard thresholds, not LLM reasoning [91, 93], so abrupt dislocations still require hard-coded controls.

8.5 Latency, Cost, and Scalability

Traditional systems win decisively below the minute scale and across broad universes; LLM agents are viable only at minute-scale or slower frequencies. Simulators resolve high-frequency trading at nanosecond granularity [18], whereas one LLM inference takes seconds [31, 118] and multi-agent pipelines take minutes, with MASS measuring 125–378 s per trading day [55] (Figure 8b). The practical floor is the minute: PACE tilts each minute over 10–60 minute windows [140], and QuantHarness degrades at 1–15 minute bars [171]. Vendor-reported typed decision models could lower this floor. TypeSafe's Jev returns a typed decision with a probability distribution and a confidence score in a reported 70–500 ms [1, 150]; if this holds for financial tasks, schema-constrained outputs would remove malformed-order errors (§9.1) and bounded components could move toward intraday frequencies. However, its calibration is untested against realized market outcomes, its training cutoff is undisclosed, no trading evaluation is published, and it forgoes the textual rationale that makes LLM theses legible.

Cost and breadth narrow the niche further: API charges accumulate with calls per decision and with breadth (Figure 8a) [55, 91, 93, 169], and where traditional systems scale linearly to thousands

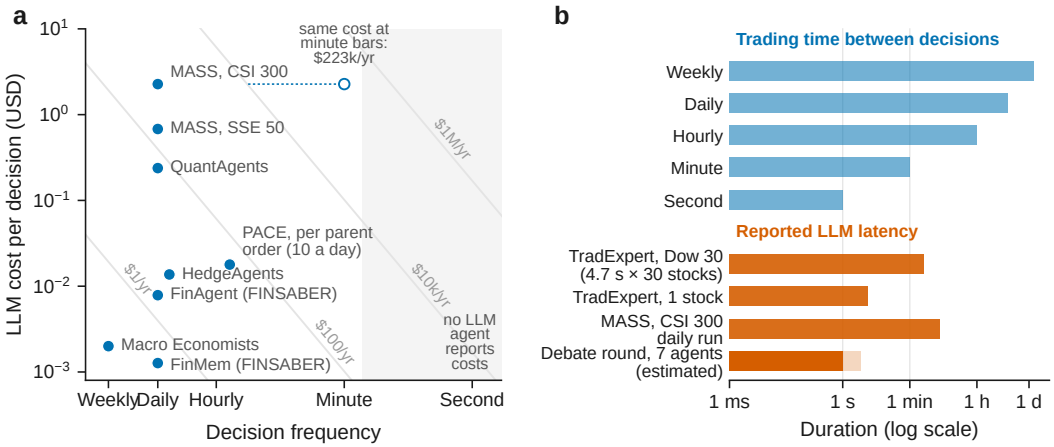


Fig. 8. Cost and latency against decision frequency. (a) Reported LLM cost per decision, with iso-lines of annual spend for one continuously running system; (b) trading time between decisions against reported or estimated LLM latency [31, 47, 55, 88, 91, 93, 118, 140].

of instruments, LLM agents hit context limits [55]. For text signals, API spend is not the binding cost; capacity, market impact, and the signal's decay relative to pipeline latency are. Lopez-Lira and Tang's signal is unprofitable at 20 bps [106], and no surveyed paper reports capacity curves or impact-adjusted returns.

8.6 Synthesis

The best-supported current hypothesis is a division of labor: *LLMs produce artifacts that a quantitative system can test and bound; quantitative systems make, size, and execute the trade.* Table 6 turns this division into deployable roles. Support for its three upstream outputs (text signals, factor and model code, and bounded execution tilts) is uneven: text signals rest on long, mostly costed samples, whereas factor and model code validated in a backtester [96] and bounded execution tilts [140] rest on single author-run studies. Within author-run studies, Alpha Miners and bounded components lead their strongest baseline in 5 of 6 studies and Traders in 6 of 11 (Table 5), too few to separate position from baseline choice. We nonetheless conjecture that the pattern is positional: the LLM's strengths (reading text, interpreting novel situations, writing legible theses and code) lie upstream of the decision, while its liabilities (memorized prices, imprecise arithmetic, unfaithful rationales, regime-blind conservatism, second-scale latency, context-bound breadth) bind at the decision itself.

Being upstream is not sufficient: the Macro Economists gain only marginally over a same-input rule, FinRL-DeepSeek's LLM signals help one policy and hurt another [10], and upstream outputs carry their own look-ahead and data-snooping risks (§7.5). LLM agents should therefore own information-intensive perception and strategy ideation, and traditional systems should own latency-sensitive execution, numerical computation, risk control, portfolio optimization, and breadth. The hypothesis would be falsified by an independently re-run autonomous trader that holds a costed edge over a passive baseline matched to its market exposure. The decisive missing test is a controlled comparison of hybrid, pure-LLM, and pure-quant architectures.

Table 7. Research agenda consolidating the open questions of §4–§9. Each row pairs an open problem with a concrete next step or measurement.

Open problem	Concrete next step or measurement	§
Hybrid vs. pure-LLM vs. pure-quant designs	One protocol (same data, window, backbone, costs, and exposure-matched baselines) comparing the three LLM paradigms with non-LLM pipelines; the decisive test of Table 6	4.2, 8.6
Value of multi-agent structure	Matched-input, multi-seed ablations on an independent benchmark	4.3
Component gains in a trading loop	Domain-specific vs. generic retrieval, layered vs. flat memory with learned forgetting, and distilled reasoning, each tested in a costed trading loop	5.1–5.5
Capacity and latency of upstream signals	Capacity curves and impact-adjusted returns; returns measured at the agent's realized decision lag	8.5
Calibrated typed decision models	Test whether vendor-reported calibration [1] holds against realized market outcomes, post-cutoff and net of costs, against LLM verbal confidence and classical classifiers	8.5
Numerical errors that reach orders	Order-level audits of how often errors and skipped tool calls reach executed orders; typed action interfaces validated deterministically against holdings and liquidity; trajectory-level extensions of CNFinBench [30] and HalluBench [80] under noisy retrieval	9.1
Attack success versus portfolio damage	Closed-loop red-teaming with mixed, adaptive, and cross-architecture attacks, scored by Net Resilient Performance (NRP) [186] extended with financial risk measures [173]	9.2
Integrity of tools and shared memory	Authenticated, provenance-tracked tool channels; memory reconciled against an independent ledger; memory-injection tests on trading agents with cross-platform memory [126]	9.2
Production readiness unreported	Staged deployment to capped real capital under the controls of §9.4, reporting monitoring, failover, and API-outage behavior alongside returns	9.3
Text-centric perception	Real-time multimodal and audio ingestion (order books, press conferences) within decision latency	9.3
Validating non-deterministic, vendor-updated models	Re-testing and re-authorization triggered by backbone updates [35, 128]; compliance modules implementing the IOSCO and FSB examination items [42, 67]; audit trails independent of possibly post-hoc reasoning traces (§8.3)	9.4
Liability and scale of autonomous trading	Liability rules for intent-free "coordination outcomes" [99, 133]; measuring how much autonomous LLM trading occurs	9.4
Monoculture and cascades	Cross-agent signal-correlation studies in securities markets varying backbone, architecture, and information set; overlap of LLM-mined factors with published and peer factors; behavior under a common backbone outage or update; LLM-populated agent-based simulations (§7.2) of the cascade to price impact	9.5
Look-ahead and version drift	Shared point-in-time datasets including delisted securities; chronologically consistent backbones whose pretraining stops before each test year [59, 135], scaled up and instruction-tuned to drive agents; reproducibility checklists (§7.5) as submission requirements	9.6
Human-agent division of labor	Joint metrics including tail errors [21, 155]; human-owned mechanical aggregation [136]; trust signals tied to informative agent confidence [174]; sign-off designs tested for automation bias [124]; autonomy calibrated to regime and position size	9.7

9 Open Challenges and Future Directions

Seven challenges block reliable production deployment of LLM trading agents. Table 7 merges them with the open questions of §4–§8 into one research agenda for the whole survey.

9.1 Hallucination and Reliability

Numerical hallucination matters most for trading because it hides in fluent reasoning and costs money once acted on. RAG and tool learning are the most effective mitigations across financial tasks [71], but grounding inherits retrieval quality: noisy knowledge-graph triplets cut detectors' Matthews correlation by 44–84% for most detector families [80], and with the filing in context GPT-4-class models still give 17–21% incorrect FinanceBench answers [68]. Inside agents the errors reach the order: StockBench agents miscalculate share quantities or emit malformed JSON actions [25]. Delegation through typed queries, schema-constrained relations, code-based calibration, or LLMs that choose only trade direction [22, 34, 58, 109] fixes the steps it covers, but every step around the tool remains a language step. **Takeaway.** Grounding and delegation shrink numerical error without eliminating it, and no surveyed paper measures how often an error reaches an executed order, because question-answering benchmarks score answers and trading benchmarks score returns.

9.2 Adversarial Robustness and Security

LLM agents add attack surfaces along the perceive–reason–act loop, from prompt injection to tool, memory, and protocol exploits [41]. Agent Security Bench reports 84.3% average success for mixed attacks against largely ineffective defenses [186], and on Web3 tasks memory injection beats prompt injection on every tested backbone, while fine-tuning on attack examples cuts its success from 85.1% to 1.7% [126]. TradeTrap [173] shows that success rates understate financial damage: a reverse-expectation prompt injection cuts an adaptive agent's Sharpe ratio from 5.72 to 0.29, tool hijacking and memory poisoning make it trade a phantom portfolio, and procedural agents lose less only because rigid pipelines cap upside as well as downside. Because attack success rates do not predict portfolio damage, generic agent benchmarks cannot certify a trading agent, and no surveyed study tests combined, adaptive, or cross-backbone attacks in a trading loop.

9.3 Engineering Challenges of Live Deployment

Continuous live operation demands engineering that published systems rarely report. Inference cost and latency confine current agents to low-frequency trading on small universes (§8.5), and multi-agent designs compound both: TradingAgents makes 11 LLM calls and 20+ tool calls per decision [169]. No surveyed study reports monitoring, failover, or degradation under API outages, and most systems remain text-centric, lacking low-latency multimodal ingestion. Because the cost, latency, data, and fault-tolerance engineering that live trading requires is largely unreported, production readiness cannot be judged from published results.

9.4 Regulatory Compliance and Legal Risks

Existing algorithmic-trading rules already cover LLM agents run by regulated firms; LLMs add uncertainty about when an algorithm has changed and what it is meant to do. EU RTS 6 requires pre-deployment testing, order and position limits, real-time monitoring, and a kill functionality [35], and US rules and guidance add pre-trade controls, change management, and a quick way to disable an algorithm [43, 152]; agents run by individuals or on crypto venues largely fall outside them. Liability is less settled. MiFID II rather than the EU AI Act governs AI trading [7], and US manipulation law requires scienter (intent), yet DRL agents have learned manipulative trading unprompted [133]. Supervisors now specify kill switches, action scope, and human-oversight ladders for agents [15, 42, 67]. Model-risk management fits least, because sampling non-determinism and silent vendor updates mean the validated configuration may not be the one running. Model-risk frameworks nonetheless translate into concrete practice: pin backbone versions and decoding settings, place prompts, memory stores, and tool schemas under change control as model components, and re-run the previously validated configuration as a challenger after each update [13, 128]. In the US, however, SR 26-2, which replaced SR 11-7 in 2026, sidesteps the issue by excluding generative and agentic AI [14]. **Takeaway.** The controls supervisors request are already law for regulated algorithmic traders; what remains unresolved is how to validate non-deterministic, vendor-updated models, who is liable for intent-free manipulation, and how AI claims to retail clients are policed [154].

9.5 Market Stability and Systemic Risk

Automated trading has already produced crises without intent: the August 2007 quant unwind through common factor positions [74], the 2010 Flash Crash [75, 151], and Knight Capital's faulty deployment, which lost over $460 million in 45 minutes [153]. Agents sharing similar LLMs could reach correlated decisions, but no study has yet shown the full cascade, and regulators find herding evidence thin [66]. Evidence on which kind of diversity matters is emerging: tacit collusion

among identical LLM pricing agents [44] falls from about 22% price elevation to 2–10% when objectives, data access, or algorithms differ, but stays near 21% under model-size heterogeneity [73]. **Takeaway.** Crowding, interacting algorithms, and faulty updates can each destabilize markets without intent; diversity in objectives, data, or algorithms decorrelates agents whereas capability asymmetry does not, but this rule comes from pricing games and stylized models and is untested in securities markets.

9.6 Reproducibility and Open-Source Ecosystem

Many systems rely on version-unstable proprietary APIs and unreplicable real-time data, and protocol reporting remains incomplete (§7.5). Open-source frameworks [102, 175, 176] and open-weight models [177] reduce API dependence, yet 74% of surveyed practitioners rate ready-to-use evaluation tools scarce or non-existent, and half call this the main bottleneck to bias mitigation [78]; public leaderboards and re-run harnesses are only beginning [129, 180]. The bottleneck has thus shifted from model access to shared evaluation tooling.

9.7 Human-AI Collaborative Trading

The evidence favors complementarity in finance because humans hold information the model lacks. Human-AI teams usually trail the better of the two unless the human alone beats the AI [155], and finance fits the exception: an ML analyst beats 54.5% of sell-side return forecasts, but humans keep the edge on illiquid, intangible-heavy, and distressed firms, and a model given analysts' forecasts avoids about 90% of their extreme errors [21]. Bank customers followed human-AI advice more often than an ML advisor's (79.2% vs. 68.3%) and earned 44.9% more, though bankers' revisions left accuracy unchanged [174]. Oversight ladders assume the human catches the agent's errors, yet users of imperfect decision aids miss unflagged errors, more so when attention is split across many agents [124]. **Takeaway.** Humans add private information and tail-error control, but trust does not track accuracy and automation bias weakens human sign-off as a safeguard; no surveyed study tests this division of labor in live trading.

10 Conclusion

This survey traced LLM trading agents from their predecessors, mapped 47 systems by capability and market (Figure 3), and audited the evidence behind their results. Four findings follow. First, independently re-run margins shrink or vanish, though the re-runs cover few systems and the original margins rest on short, noisy windows (§7.4, §7.5). Second, the best-supported current hypothesis places the LLM upstream of the trade, in text signals, factor and model code, and bounded execution tilts (§8.6). Third, controlled comparisons do not establish a multi-agent advantage (§4.3). Fourth, structure and input design matter alongside the backbone, and the best structure depends on the backbone; no study varies scale with structure held fixed (§4.3, §5.2).

Table 6 turns this division of labor into controls for practitioners, giving each LLM role a deterministic gate, a bound on its action, and a monitoring target. Table 7 lists the open gaps, including unmeasured numerical errors in executed orders and the scarcity of live, real-capital evaluation [187]. Until a backtested Sharpe ratio above the passive baseline survives a bias-mitigated re-run, it remains exploratory.

Acknowledgments

This research is partially supported by the National Research Foundation, Singapore, and the Cyber Security Agency under its National Cybersecurity R&D Programme (NCRP25-P04-TAICeN), and by Ripple under the University Blockchain Research Initiative (UBRI) [40]. Any opinions, findings and conclusions or recommendations expressed in this material are those of the author(s) and

do not reflect the views of National Research Foundation, Singapore, Cyber Security Agency of Singapore, and Ripple.

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